EXPLORING CUSTOMER TRANSACTIONS USING R

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#Project Overview In this project i will use R to perform a basic data science workflow on customer transaction dataset (e-commerce). The project introduces the use of R for data import,data inspection,data cleaning, exploretory data analysis(EDA),visualizationm and interpretation. The dataset contains transaction-level information including customer demographics,product characteristics,payment methods,transaction dates,prices,quantities,ratings and membership status.It help us identify simple patterns that can support better business decisions.

#Project objectives The project aims to achieve the following objectives 1.Import a dataset in R. 2.Inspect the structure and characteristics of data. 3.Identify and handle missing or inconsistent values. 4.Convert variables to appropriate data types. 5.Generate descriptive statistics using R. 6.Create basic statistical and graphical summaries. 7.Explore relationships between variables. 8.Communicate data-driven findings.

#Setting a working directory

```
setwd("C:/Users/Administrator/Desktop/PROFFESIONAL TRAINING")
```

#import libraries

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(sjmisc)
library(rio)
library(stats)
library(forecast)
library(summarytools)
library(ggthemes)
library(formatR)
library(tidyr)
library(kableExtra)
library(psych)
library(plotly)
```

#import dataset

```
comm_data<-read.csv("comm.csv")
```

#display first observations

```
head(comm_data)
```

```
  TransactionID CustomerID ProductID Quantity    PaymentMethod TransactionDate
1             1        207        14        5              UPI      12/28/2023
2             2        253        15        1 Cash on Delivery       4/17/2023
3             3        110        19        5 Cash on Delivery       1/17/2023
4             4        256         1        3       Debit Card      10/23/2023
5             5        274         2        5       Debit Card       6/30/2023
6             6         52        15        2      Credit Card       6/14/2023
  ProductCategory  Price Rating TotalAmount Age Gender    Location
1        Clothing  33.36    4.2      166.80  50   Male     Houston
2     Electronics 389.03    2.2      389.03  19 Female     Chicago
3        Clothing 145.89    1.7      729.45  37 Female     Houston
4 Home Appliances 215.02    2.7      645.06  50   Male Los Angeles
5 Home Appliances 325.48    3.4     1627.40  24   Male    New York
6     Electronics 389.03    2.2      778.06  50   Male     Houston
  MembershipStatus
1            Basic
2          Premium
3            Basic
4          Premium
5          Premium
6          Premium
```

#display last observations

```
tail(comm_data)
```

```
    TransactionID CustomerID ProductID Quantity    PaymentMethod
295           295        181         9        5 Cash on Delivery
296           296        157        15        5 Cash on Delivery
297           297        293        13        5       Debit Card
298           298        189        18        4       Debit Card
299           299        182        15        2       Debit Card
300           300        183        12        3       Debit Card
    TransactionDate ProductCategory  Price Rating TotalAmount Age Gender
295       9/28/2023     Electronics 132.51    2.3      662.55  30   Male
296       3/13/2023     Electronics 389.03    2.2     1945.15  57   Male
297       10/5/2023        Clothing 243.41    5.0     1217.05  59   Male
298        8/2/2023 Home Appliances 341.40    3.0     1365.60  31   Male
299      11/12/2023     Electronics 389.03    2.2      778.06  63 Female
300       2/19/2023 Home Appliances 466.49    2.6     1399.47  23 Female
       Location MembershipStatus
295 Los Angeles          Premium
296     Chicago             Gold
297 Los Angeles            Basic
298 Los Angeles             Gold
299     Houston            Basic
300     Chicago            Basic
```

#number of observations and variables

```
dim(comm_data)
```

```
[1] 300  14
```

#structure and characteristics of data

```
str(comm_data)
```

```
'data.frame':   300 obs. of  14 variables:
 $ TransactionID   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ CustomerID      : int  207 253 110 256 274 52 191 165 18 169 ...
 $ ProductID       : int  14 15 19 1 2 15 10 5 5 17 ...
 $ Quantity        : int  5 1 5 3 5 2 1 5 4 2 ...
 $ PaymentMethod   : chr  "UPI" "Cash on Delivery" "Cash on Delivery" "Debit Card" ...
 $ TransactionDate : chr  "12/28/2023" "4/17/2023" "1/17/2023" "10/23/2023" ...
 $ ProductCategory : chr  "Clothing" "Electronics" "Clothing" "Home Appliances" ...
 $ Price           : num  33.4 389 145.9 215 325.5 ...
 $ Rating          : num  4.2 2.2 1.7 2.7 3.4 2.2 3.7 1.4 1.4 1.9 ...
 $ TotalAmount     : num  167 389 729 645 1627 ...
 $ Age             : int  50 19 37 50 24 50 40 37 25 53 ...
 $ Gender          : chr  "Male" "Female" "Female" "Male" ...
 $ Location        : chr  "Houston" "Chicago" "Houston" "Los Angeles" ...
 $ MembershipStatus: chr  "Basic" "Premium" "Basic" "Premium" ...
```

#summary statistics

```
summary(comm_data)
```

```
 TransactionID      CustomerID      ProductID        Quantity    
 Min.   :  1.00   Min.   :  4.0   Min.   : 1.00   Min.   :1.000  
 1st Qu.: 75.75   1st Qu.: 77.0   1st Qu.: 6.00   1st Qu.:2.000  
 Median :150.50   Median :156.5   Median :11.00   Median :3.000  
 Mean   :150.50   Mean   :154.8   Mean   :10.66   Mean   :3.127  
 3rd Qu.:225.25   3rd Qu.:229.2   3rd Qu.:15.00   3rd Qu.:5.000  
 Max.   :300.00   Max.   :299.0   Max.   :20.00   Max.   :5.000  
                                                                 
   PaymentMethod  TransactionDate  ProductCategory     Price       
 Length   :300   Length   :300    Length   :300    Min.   : 33.36  
 N.unique :  4   N.unique :216    N.unique :  3    1st Qu.:145.89  
 N.blank  :  0   N.blank  :  0    N.blank  :  0    Median :215.02  
 Min.nchar:  3   Min.nchar:  8    Min.nchar:  8    Mean   :253.52  
 Max.nchar: 16   Max.nchar: 10    Max.nchar: 15    3rd Qu.:350.44  
                                                   Max.   :466.49  
                                                                   
     Rating       TotalAmount           Age             Gender     
 Min.   :1.400   Min.   :  33.36   Min.   :18.00   Length   :300  
 1st Qu.:2.100   1st Qu.: 350.44   1st Qu.:30.00   N.unique :  3  
 Median :2.600   Median : 662.55   Median :42.00   N.blank  : 13  
 Mean   :2.928   Mean   : 803.12   Mean   :41.79   Min.nchar:  0  
 3rd Qu.:3.700   3rd Qu.:1134.50   3rd Qu.:52.25   Max.nchar:  6  
 Max.   :5.000   Max.   :2332.45   Max.   :65.00                  
                                   NAs    :8                      
      Location    MembershipStatus
 Length   :300   Length   :300    
 N.unique :  4   N.unique :  3    
 N.blank  :  0   N.blank  :  0    
 Min.nchar:  7   Min.nchar:  4    
 Max.nchar: 11   Max.nchar:  7    
```

#identify and handle missing values

```
miss_value<-sum(is.na(comm_data))
head(miss_value)
```

```
[1] 8
```

#identify missing colms

```
colSums(is.na(comm_data))
```

```
   TransactionID       CustomerID        ProductID         Quantity 
               0                0                0                0 
   PaymentMethod  TransactionDate  ProductCategory            Price 
               0                0                0                0 
          Rating      TotalAmount              Age           Gender 
               0                0                8                0 
        Location MembershipStatus 
               0                0 
```

#identify duplicate records

```
sum(duplicated(comm_data))
```

```
[1] 0
```

#Replace missing values with mean

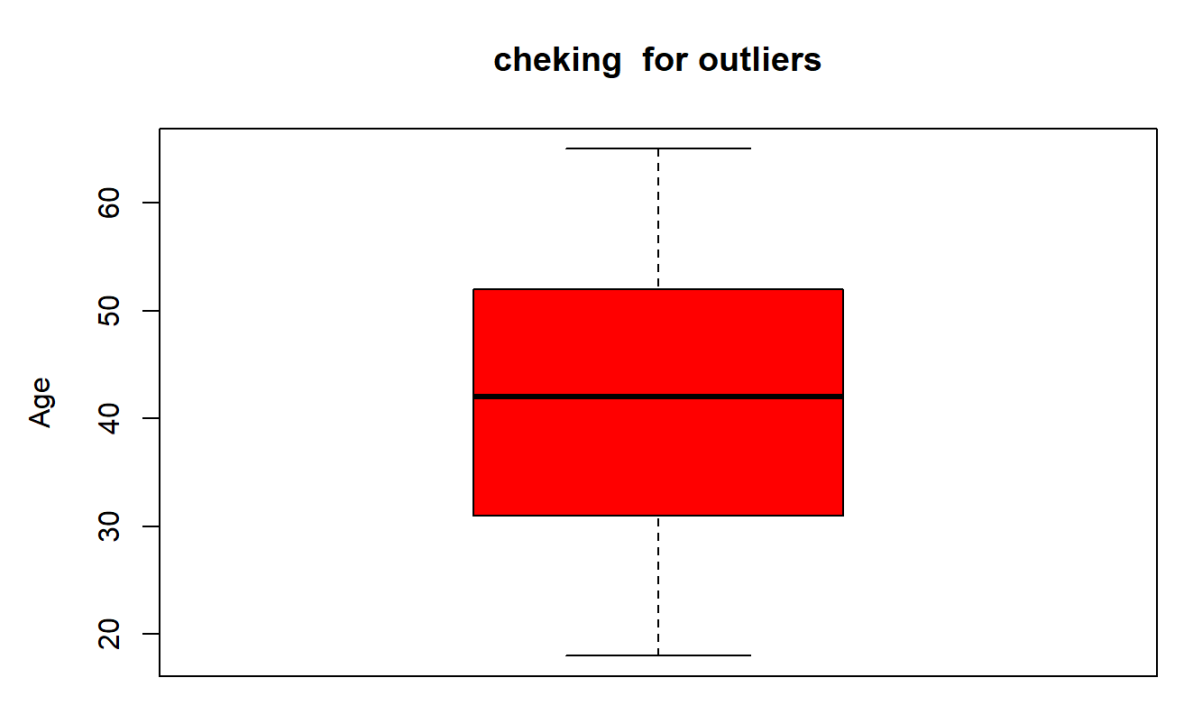
```
comm_data$Age[is.na(comm_data$Age)]<-mean(comm_data$Age,na.rm=TRUE)
```

```
colSums(is.na(comm_data))
```

```
   TransactionID       CustomerID        ProductID         Quantity 
               0                0                0                0 
   PaymentMethod  TransactionDate  ProductCategory            Price 
               0                0                0                0 
          Rating      TotalAmount              Age           Gender 
               0                0                0                0 
        Location MembershipStatus 
               0                0 
```

#Checking for outliers

```
boxplot(comm_data$Age,
        ylab="Age",
        main="cheking  for outliers",
        col="red")
```



#descriptive statistics

```
describe(comm_data)
```

```
                  vars   n   mean     sd median trimmed    mad   min     max
TransactionID        1 300 150.50  86.75 150.50  150.50 111.19  1.00  300.00
CustomerID           2 300 154.83  85.38 156.50  155.66 111.94  4.00  299.00
ProductID            3 300  10.66   5.58  11.00   10.73   7.41  1.00   20.00
Quantity             4 300   3.13   1.46   3.00    3.16   1.48  1.00    5.00
PaymentMethod*       5 300   2.55   1.18   3.00    2.56   1.48  1.00    4.00
TransactionDate*     6 300 109.66  62.24 110.50  110.10  80.80  1.00  216.00
ProductCategory*     7 300   1.96   0.82   2.00    1.95   1.48  1.00    3.00
Price                8 300 253.52 120.62 215.02  250.44 135.49 33.36  466.49
Rating               9 300   2.93   1.06   2.60    2.85   1.04  1.40    5.00
TotalAmount         10 300 803.12 568.91 662.55  745.60 565.82 33.36 2332.45
Age                 11 300  41.79  13.10  42.00   41.74  16.31 18.00   65.00
Gender*             12 300   2.49   0.58   3.00    2.54   0.00  1.00    3.00
Location*           13 300   2.37   1.14   2.00    2.33   1.48  1.00    4.00
MembershipStatus*   14 300   1.98   0.83   2.00    1.98   1.48  1.00    3.00
                   range  skew kurtosis    se
TransactionID     299.00  0.00    -1.21  5.01
CustomerID        295.00 -0.07    -1.23  4.93
ProductID          19.00 -0.09    -1.17  0.32
Quantity            4.00 -0.10    -1.36  0.08
PaymentMethod*      3.00 -0.06    -1.49  0.07
TransactionDate*  215.00 -0.05    -1.22  3.59
ProductCategory*    2.00  0.07    -1.51  0.05
Price             433.13  0.18    -0.96  6.96
Rating              3.60  0.57    -0.73  0.06
TotalAmount      2299.09  0.78    -0.32 32.85
Age                47.00 -0.01    -1.08  0.76
Gender*             2.00 -0.61    -0.61  0.03
Location*           3.00  0.17    -1.40  0.07
MembershipStatus*   2.00  0.03    -1.55  0.05
```

```
head(describe)
```

```
1 function (x, na.rm = TRUE, interp = FALSE, skew = TRUE, ranges = TRUE, 
2     trim = 0.1, type = 3, check = TRUE, fast = NULL, quant = NULL, 
3     IQR = FALSE, omit = FALSE, data = NULL, size = 50)            
4 {                                                                  
5     if (inherits(x, "formula")) {                                  
6         ps <- fparse(x)                                            
```

#Data preparation #convert categorical variables to factors

```
comm_data$PaymentMethod<-as.factor(comm_data$PaymentMethod)
comm_data$Gender<-as.factor(comm_data$Gender)
comm_data$Location<-as.factor(comm_data$Location)
comm_data$ProductCategory<-as.factor(comm_data$ProductCategory)
comm_data$MembershipStatus<-as.factor(comm_data$MembershipStatus)
```

#convert transaction date to proper date format

```
comm_data$TransactionDate<-as.Date(comm_data$TransactionDate,
                                   format = "%m%d%y")
```

#create the cleaned analytical dataset

```
clean_data<-na.omit(comm_data)
```

```
dim(clean_data)
```

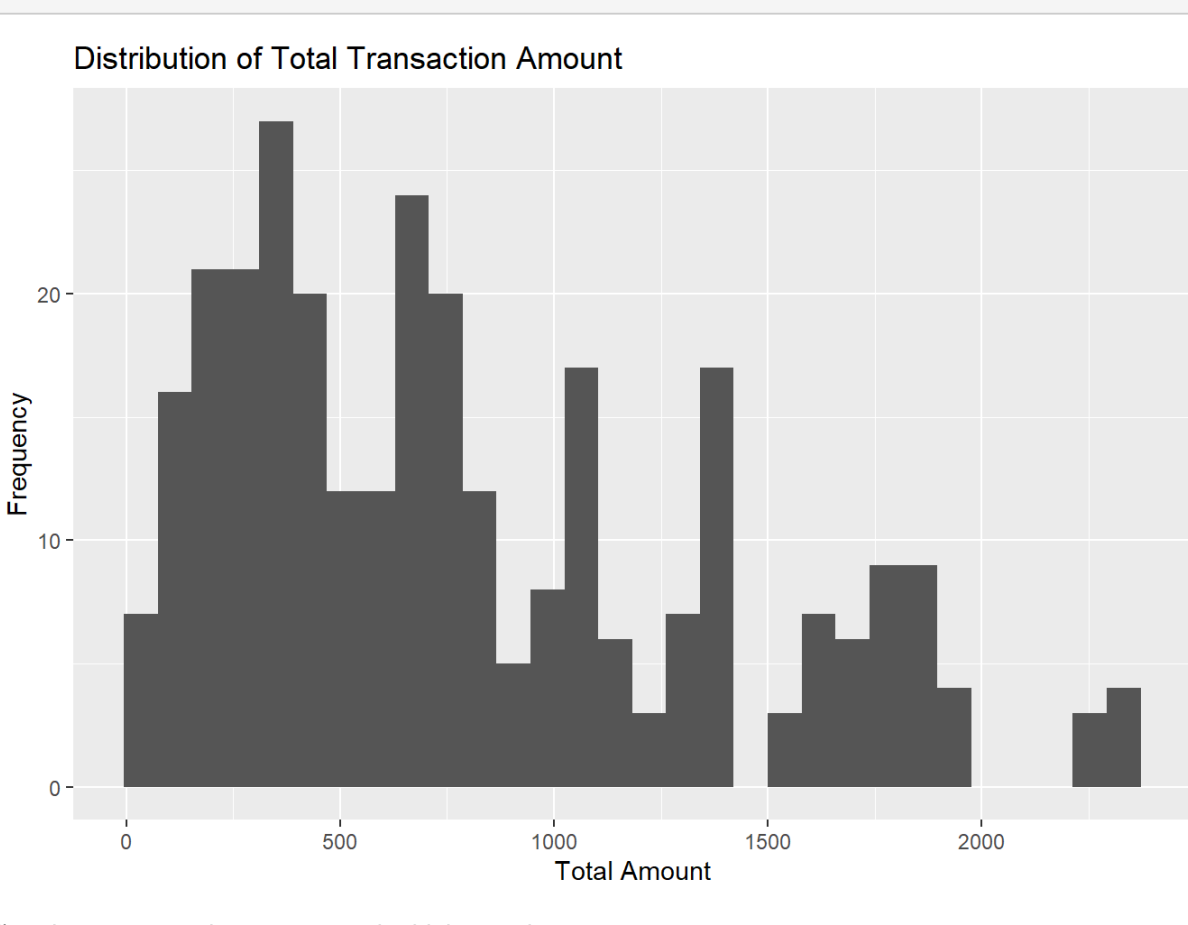
```
[1]  0 14
```

#exploratory data analysis #distribution of total amount

```
summary(comm_data$TotalAmount)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  33.36  350.44  662.55  803.12 1134.50 2332.45 
```

```
ggplot(comm_data,aes(x=TotalAmount))+
  geom_histogram(bins=30)+
  labs(title="Distribution of Total Transaction Amount",
       x="Total Amount",
       y="Frequency")
```



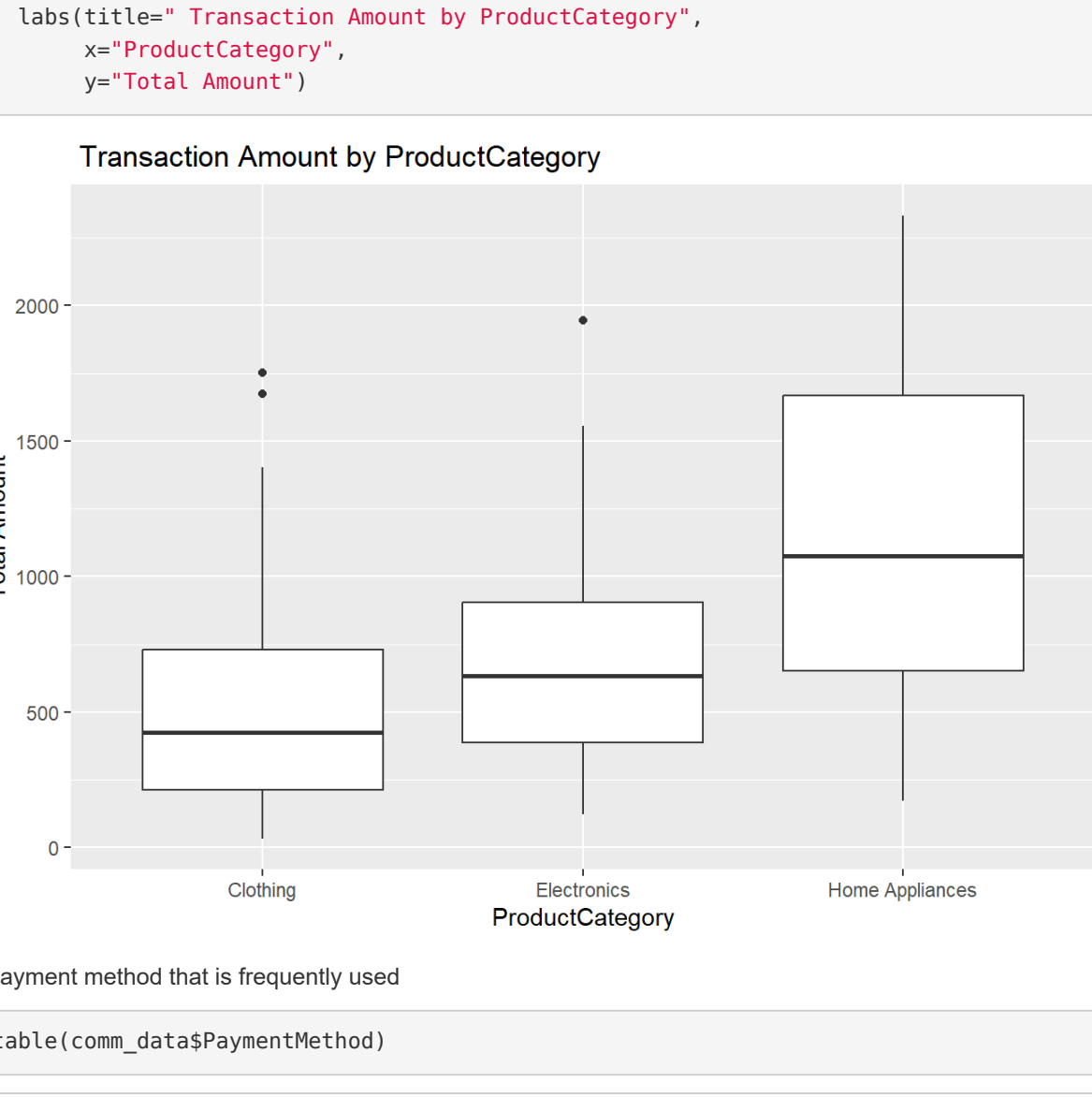
#product category that generates the highest sales

```
aggregate(TotalAmount~ProductCategory,
          data=comm_data,sum)
```

```
  ProductCategory TotalAmount
1        Clothing    60536.40
2     Electronics    69501.41
3 Home Appliances   110897.21
```

#Home appliances therefore generate the highest total sales of 110897.21

```
ggplot(comm_data,aes(x=ProductCategory,y=TotalAmount))+
  geom_boxplot()+
  labs(title=" Transaction Amount by ProductCategory",
       x="ProductCategory",
       y="Total Amount")
```



#payment method that is frequently used

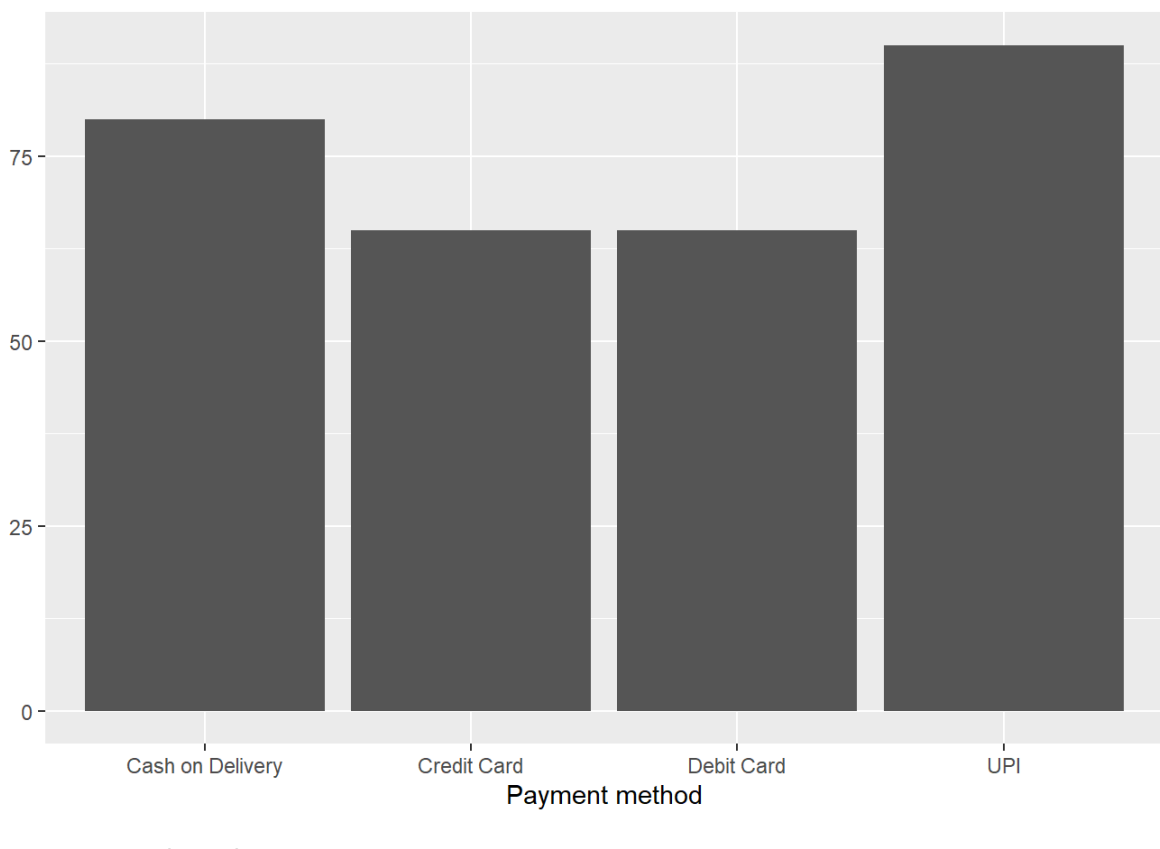
```
table(comm_data$PaymentMethod)
```

```

Cash on Delivery      Credit Card       Debit Card              UPI 
              80               65               65               90 
```

#UPI is therefore the most frequently used payment method with 90 transactions

```
ggplot(comm_data,aes(x=PaymentMethod))+
  geom_bar()+
  labs(title="Frequency of payment methods",
       x="Payment method",
       y="Number of transactions")
```



#average transaction value

```
mean(comm_data$TotalAmount)
```

```
[1] 803.1167
```

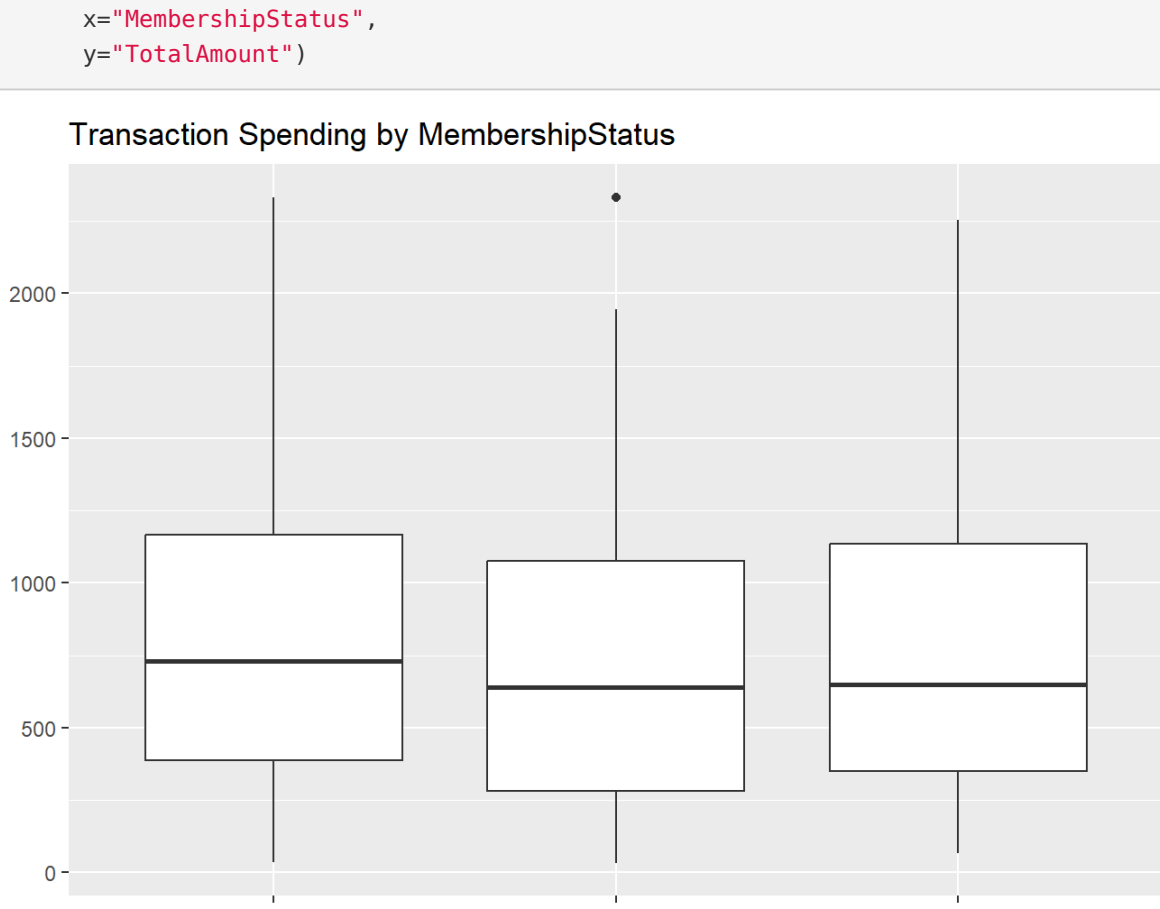
#transaction spending between membershipstatus

```
aggregate(TotalAmount~MembershipStatus,
          data=comm_data,mean)
```

```
  MembershipStatus TotalAmount
1            Basic    827.7787
2             Gold    759.8215
3          Premium    818.3522
```

#The average transaction amount differs between membershipstatus, basic members have the highest transaction followed by premium then gold being the lowest

```
ggplot(comm_data,aes(x=MembershipStatus,y=TotalAmount))+
  geom_boxplot()+
  labs(title="Transaction Spending by MembershipStatus",
       x="MembershipStatus",
       y="TotalAmount")
```



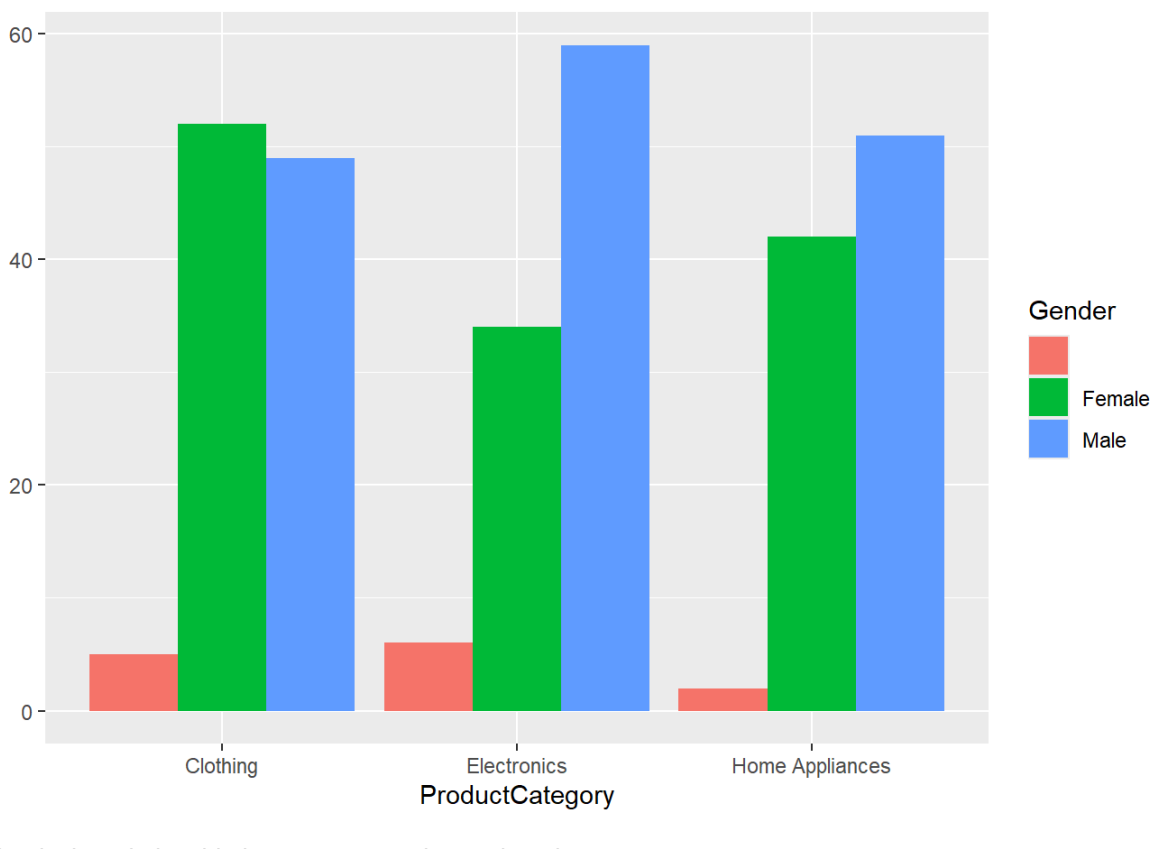
#is there any association between gender and product category

```
table(comm_data$Gender,comm_data$ProductCategory)
```

```
        
         Clothing Electronics Home Appliances
                5           6               2
  Female       52          34              42
  Male         49          59              51
```

#there is no association between gender and product category

```
ggplot(comm_data,aes(x=ProductCategory,fill=Gender))+
  geom_bar(position = "dodge")+
  labs(title="ProductCategory by Gender",
       x="ProductCategory",
       y="Number of Transactions")
```



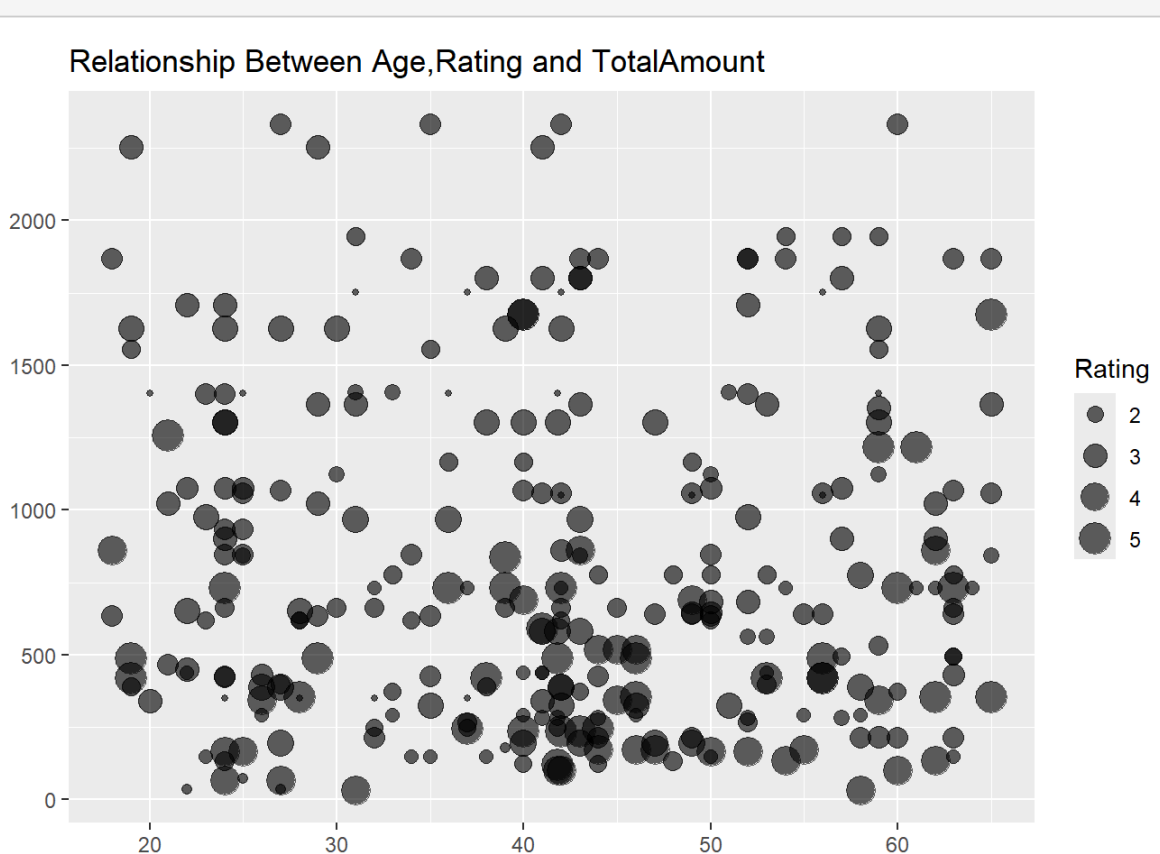
#what is the relationship between age,rating and total amount

```
cor(comm_data[c("Age","Rating","TotalAmount")],
    use = "complete.obs")
```

```
                    Age      Rating TotalAmount
Age          1.00000000  0.03805788 -0.01609505
Rating       0.03805788  1.00000000 -0.12783771
TotalAmount -0.01609505 -0.12783771  1.00000000
```

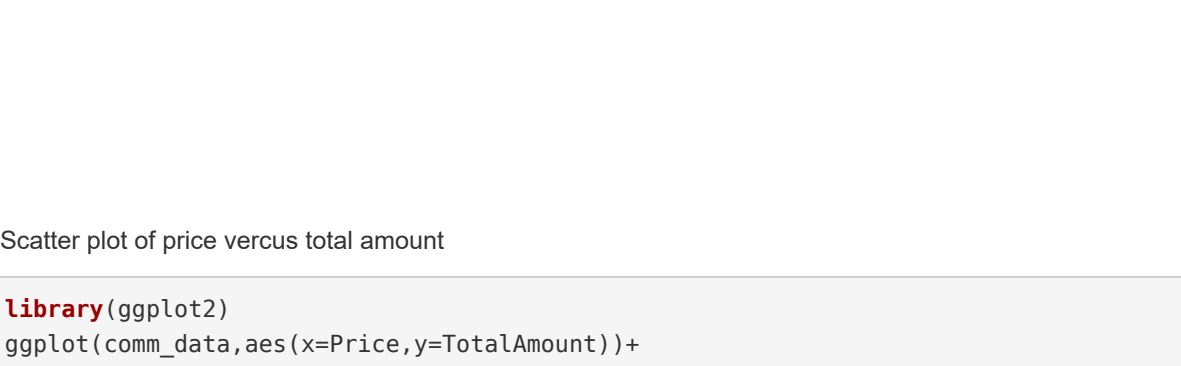
#the relationship between age and rating is positively weak ,between age and total amount is negatively weak,between rating and age positively weak,between rating and total amount is negatively weak,between total amount and age is negatively weak and between total aamount and rating is also negatively weak.

```
ggplot(comm_data,aes(x=Age,y=TotalAmount,size=Rating))+
  geom_point(alpha = 0.6)+
  labs(title="Relationship Between Age,Rating and TotalAmount",
       x="Age",
       y="TotalAmount",
       size="Rating")
```



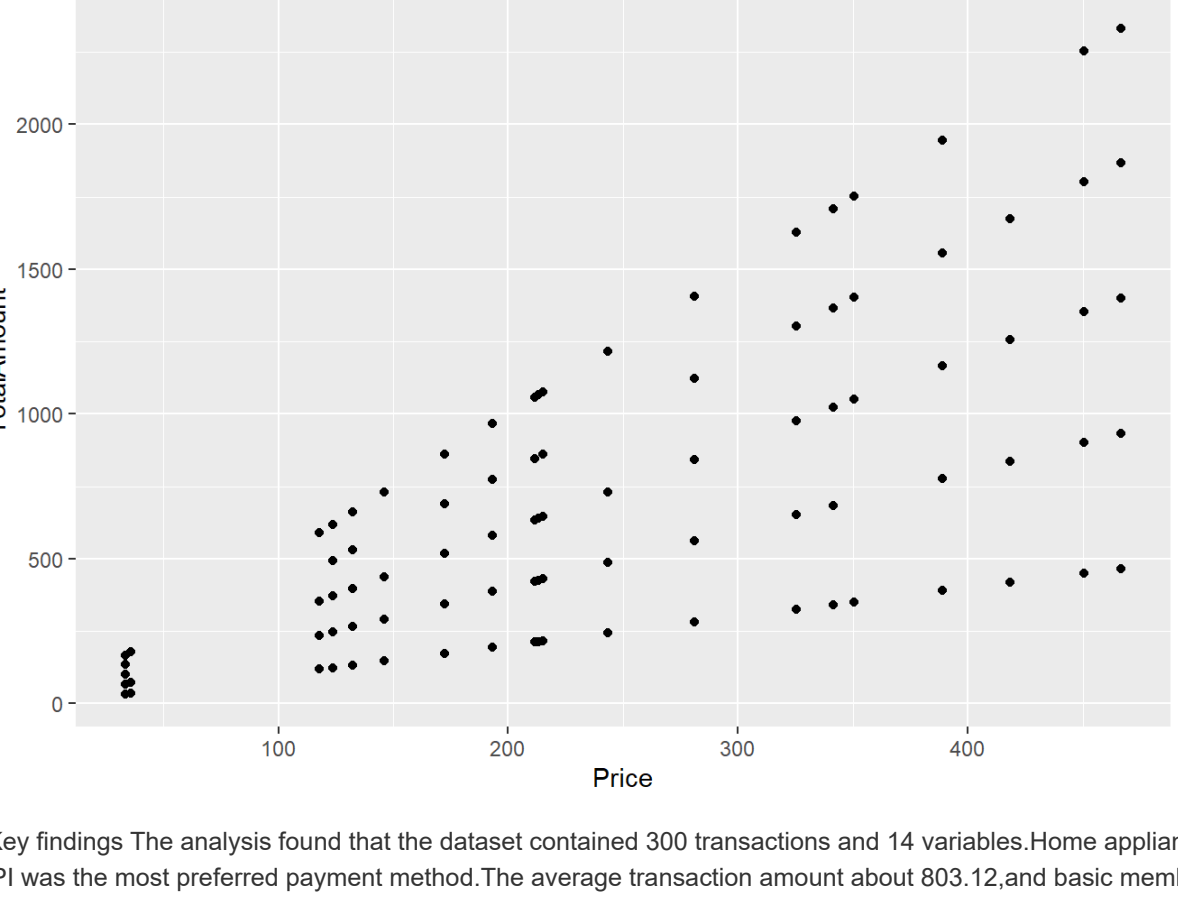
#creat a 3D scatter plot

```
plot_ly(
  data=comm_data,
  x=~Age,
  y=~Rating,
  z=~TotalAmount,
  type="scatter3d",
  mode="markers",
  marker=list(size=5)
)%>%
  layout(
    title="Relationship Between Age,Rating and TotalAmount",
    scene=list(
      xaxis=list(title="Age"),
      yaxis=list(title="Rating"),
      zaxis=list(title="TotalAmount")
    )
  )
```



#Scatter plot of price vercus total amount

```
library(ggplot2)
ggplot(comm_data,aes(x=Price,y=TotalAmount))+
  geom_point()+
  labs(title="Price versus TotalAmount",
       x="Price",
       y="TotalAmount")
```



#Key findings The analysis found that the dataset contained 300 transactions and 14 variables.Home appliances had the highest total sales,while UPI was the most preferred payment method.The average transaction amount about 803.12,and basic members had the highest average spending.There was very weak relationship between age and spending,while gender and product category showed no strong relationship.After cleaning the data,287 complete records remained.

#conclusion The transaction data helps us understand how customers shop,it shows what products sell most,how much customers spend,which payment methods they prefer and how spending differs by membership status.In our data,Home Appliances had the highest sales,UPI was the most used payment method and basic members had the highest average spending.